FIRESIDE CHAT · PREPARED FOR THE PLATFORM TEAM · DATA ENGINEER

Nexus, **for Abound.**

Every change in Aurora, in the lake exactly once. Proven against the source. Ready for model calibration.

NEXUS is the platform I built and ran at Lifepal, a regulated insurance marketplace: CDC from PostgreSQL and MySQL into an S3 lake with AWS DMS and Debezium, Spark and Glue merges that are idempotent under replay, and the feature pipelines behind risk and fraud models. Here it is applied to Abound's stack, Aurora → DMS → Glue → Iceberg → Athena: every change keyed on its change sequence, every delete a tombstone, every table reconciled to the source, and calibration data built as of the decision. Everything below runs at **rfim.github.io/nexus-abound**.

Aurora · DMS · Glue · Spark · Iceberg · Athena

BUILT IN lifepal → PRESENTED TO ABOUND

THE PATH

what runs in the repository



Op, cdc_ts and AR_H_CHANGE_SEQ on every row.

Every change keyed on (key, change_seq).

Deletes are tombstones; a reload is authoritative.

Reconciliation caught 16 rows DMS never sent.

90% less scanned for a per-customer read.

PROVEN AT LIFEPAL **24h → <5m** reporting latency **6h → 90m** nightly batch **$11k** vs $19k projection **−60%** ad-hoc requests **500+** checks in CI **−60%** data-trust escalations

SIX CDC PROBLEMS

planted, handled, tested

PROBLEM	HANDLED
Restart re-send	14 changes in a new file → skipped in bronze
Late file	2026-09-03 file landed 2026-09-18 → ordered by change_seq
Delete, then older update	tombstone, so no resurrection
ALTER TABLE ADD COLUMN	metadata-only ADD COLUMNS in bronze
Same commit timestamp	53 two-statement transactions → change_seq
Changes never delivered	16 changes → reconciliation + reload

FOR DATA SCIENCE

calibration, point in time

SCORE BAND	LOANS	PREDICTED	OBSERVED
0–10%	59	7%	8%
10–20%	164	15%	19%
20%+	98	25%	30%

Point in time: the naive income feature differs for 72 of 111 decisions, by £2,807 on average.

IF I JOINED THE TEAM

the first 90 days

1 Days 1–30 · Listen and ship
Learn the lake, DMS tasks and models from their owners. Ship a small fix early.

2 Days 31–60 · Prove it
Reconciliation from Aurora for the tables Data Science uses; change-volume alerts.

3 Days 61–90 · Move one workload
Model calibration on the lake, point in time; one RDS read switched off.

4 Day 90 · Measured
Fresher data, cheaper queries, less load on Aurora.

THE ROLE, FEATURE BY FEATURE

built in Lifepal · applied here

FEATURE	AT LIFEPAL	HERE
AWS DMS · CDC	DMS and Debezium from PostgreSQL and MySQL into S3	bronze.py: keyed on (key, change_seq)
Spark · Glue	Moved reporting, analytics and training off RDS	glue/cdc_job.py on Glue 5.0
Iceberg	Delta in production; the ideas map one to one	MERGE, hidden partitions, sort order, compaction
Athena · cost	$11k a month against a $19k projection	90% and 77% less scanned
Governance	500+ checks, PII masking, RBAC, lineage	reconciliation, quality rules, erasure
Lambda · ECS · Fargate	Services on Lambda and ECS	events, reloads and backfills around the job

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